NEDA KEYHANVAR, PhD

Postdoctoral Researcher

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PROFESSIONAL SUMMARY

Translational scientist with a PhD in Medical Biotechnology and postdoctoral training at the University of California, San Francisco, specializing in mammalian cell and stem cell engineering, functional genomics, CRISPR-based screening, and aging biology. Over ten years of experience independently designing and executing multidisciplinary research programs spanning transgenic mouse models, epitranscriptomics, mechanobiology, and multi-omics integration, with a consistent record of establishing novel experimental platforms and translating mechanistic findings toward publication. Demonstrated expertise in expression vector design, recombinant protein production, CRISPR/Cas9 genome engineering, and high-throughput functional screening, complemented by strong scientific writing, mentorship, and cross-disciplinary collaboration skills. Seeking a research position at an academic institute where rigorous, innovative science advances fundamental understanding of aging, stem cell biology, and disease mechanisms.

EDUCATION

PhD, Medical Biotechnology, Tabriz University of Medical Sciences, Iran. 2014 – 2021
MSc, Medical Biotechnology, Golestan University of Medical Sciences, Iran. 2010 – 2013
BSc, Cell & Molecular Biology, Azerbaijan University of Tarbiat Moallem, Iran. 2005 – 2009

RESEARCH EXPERIENCE

Postdoctoral Researcher | UCSF – Martens Lab | San Francisco, CA **Feb 2025 – Present**

- Lead an independent research program investigating how aging-associated extracellular matrix (ECM) remodeling drives changes in tissue mechanics, inflammatory signaling, and cellular mechanoprotection in the thymus, establishing a new mechanistic framework for age-related immune decline.
- Engineered and characterized a conditional knockout transgenic mouse model demonstrating that YAP/TAZ ablation triggers thymic involution within 19 days versus 8-9 months under natural aging, establishing an accelerated preclinical aging platform that substantially reduces experimental timelines and cost.
- Independently pioneered the lab's first decellularized thymic ECM scaffold platform across five age timepoints, establishing a reproducible mechanobiology pipeline and opening a new research program within the group.
- Designed and implemented an end-to-end, NGS-based multi-omics pipeline spanning primary cell isolation, multi-color FACS sorting, sample preparation for bulk RNA-seq, and single-cell RNA-seq, supporting multiple concurrent projects.
- Collaborate closely with a computational postdoctoral fellow to integrate mechanobiology data with single-cell multi-omics and spatial transcriptomics using systems-biology approaches, expanding the laboratory's cross-disciplinary analytical capacity.
- Serve as the laboratory's sole *in vivo* specialist, independently managing all transgenic mouse work including IACUC protocol preparation, colony management, breeding, genotyping, and over 20 sterile survival surgeries (intrathymic injection with and without thoracotomy) with high reproducibility across multiple transgenic lines compliant with controlled substances regulation.
- Perform multi-color immunophenotyping of thymic stromal and immune cell populations by flow cytometry to characterize the effects of YAP/TAZ loss on immune cell composition and T-cell development.
- Maintain all experimental documentation to Good Laboratory Practice (GLP) standards using Benchling ELN, ensuring data integrity and document control consistent with ISO compliance frameworks.
- Selected for an oral presentation and poster at the Bakar Aging Research Institute (BARI) Symposium (2025), communicating findings to a multidisciplinary audience of aging researchers and clinicians.
- Mentor a research associate through 18 months of training, culminating in five graduate school interviews and admission to a PhD program; additionally supervise an undergraduate researcher and a rotating PhD student.

Postdoctoral Researcher | UCSF – Vissers Lab | San Francisco, CA **Oct 2023 – Jan 2025**

- Independently led a project engineering CRISPR/Cas9 knock-in hiPSC reporter cell lines using a novel split-GFP (GFP11-only) strategy, achieving the first endogenous tagging of m6A RNA methylation machinery in the laboratory and establishing epitranscriptomic reporter methodologies not previously available (first-author manuscript in preparation).
- Designed and executed the laboratory's first pooled CRISPRi screen using a 12,000-gRNA library across 150 million sorted cells, identifying upstream regulators of epitranscriptomic networks and validating hits through neural progenitor cell (NPC) differentiation and hypoxia functional assays.
- Established lentiviral vector production (HEK293T), transduction, transfection, and nucleofection workflows for mammalian cell engineering; applied FACS-based high-throughput clone screening, selection, and characterization to support stable cell line development and functional genomics studies.
- Performed RT-qPCR for gene expression analysis and target validation across multiple projects, including validation of CRISPRi screening hits and stem cell differentiation studies.
- Drafted comprehensive lab reports, experimental protocols, and technical conclusions used for Design History Files (DHFs), and presented data findings during multi-functional group meetings.
- Provided sustained experimental and troubleshooting support (20+ structured sessions) for a brain organoid project, contributing to co-authorship on a published preprint identifying human-specific roles of METTL5 in neurodevelopment via CHCHD2 regulation (Turkalj et al., bioRxiv 2025).
- Mentored a research technician through daily hands-on scientific guidance and supported his graduate school preparation via CV development, interview coaching, and literature review, resulting in a successful application.

Research Scientist | Stem Cell and Regenerative Medicine Institute | Tabriz, Iran **Sep 2014 – Sep 2023**

- Built and maintained a four-cell-type primary biobank (mesenchymal stem cells, fibroblasts, keratinocytes, adipose-derived stromal cells) supporting translational regenerative medicine programs.
- Advanced stem cell-based tissue engineering strategies for wound healing and skin regeneration, contributing to preclinical research pipelines with direct translational potential.
- Co-authored five or more peer-reviewed publications and served as an invited peer reviewer for the journal BioImpacts (6 manuscripts), reflecting recognized expertise in stem cell and regenerative medicine research.
- Trained and mentored 15+ graduate and undergraduate students in experimental design, laboratory technique, data analysis, and scientific communication, strengthening institutional research capacity.

Graduate Researcher | Tabriz University of Medical Sciences, Tabriz, Iran | **Oct 2014 – Sep 2021**

- Independently designed and engineered a novel thermoresponsive polymer scaffold directing adipose-derived stem cell differentiation into keratinocyte-like cell sheets, developed end-to-end from biomaterial design through functional validation ("KLC Sheet").
- Designed expression vectors, cloned recombinant human hepcidin-25 constructs, and optimized recombinant protein expression in baculovirus insect cell systems, including protein characterization and functional validation.
- Established robust protocols for stem cell culture, biomaterial fabrication, and scaffold characterization, forming the methodological foundation for the lab's regenerative medicine research program.
- Produced a strong doctoral publication record of 3 first-author and 5 co-author peer-reviewed papers and represented the laboratory at 2 international conferences.

PUBLICATIONS

(Reverse chronological; complete list available via Google Scholar and ORCID: 0000-0001-6852-9865)

1. **Keyhanvar N,** Waltier-Soriano C, Leonetti M, Kampmann M, Vissers C. Upstream regulators of key epitranscriptomic enzymes contextualize m6A dynamics. Expected submission: Dec 2026. (**Under preparation**)
2. **Keyhanvar N**, Biancardi F, Zhurikhina A, Jang S, Martens HL. Multi-organ collection from mouse models for aging studies. JoVE. 2025;(226):e69128. (**co-first author**)
3. Turkalj EM, Kuljanishvili M, Kang G, Liu I, Ghent C, Waltier Soriano C, **Keyhanvar N**, Brody D, Oldham M, Vissers C. Cortical organoids reveal human-specific roles of METTL5 in neurodevelopment via regulation of CHCHD2. *bioRxiv*. 2025.
4. Adel M, Keyhanvar P, Zahmatkeshan M, Bayandori M, Teimourian S, Hooshyar S, **Keyhanvar N**. Boron nitride nanotubes: Unlocking a new frontier in biomedicine. *BioNanoScience*. 2025;15(2):226.

5. Raoufinia R, Rahimi HR, **Keyhanvar N**, Moghbeli M, Abdyazdani N, Rostami M, Naghipoor K, Forouzanfar F, Foroudi S, Saburi E. Advances in treatments for epidermolysis bullosa (EB): emphasis on stem cell-based therapy. *Stem Cell Reviews and Reports*. 2024;20(5):1200–1212.
6. Raoufinia R, Arabnezhad A, **Keyhanvar N**, Abdyazdani N, Saburi E, Naseri N, Niazi F, Niazi F, Namdar AB, Rahimi HR. Leveraging stem cells to combat hepatitis: a comprehensive review of recent studies. *Molecular Biology Reports*. 2024;51(1):459.
7. Adel M, Keyhanvar P, Zahmatkeshan M, Tavangari Z, **Keyhanvar N**. A comparative simulation study of piezoelectric properties in zigzag and armchair boron nitride nanotubes: by discovering a pioneering protocol. *Journal of Mathematical Chemistry*. 2024;62(10):2943–2958.
8. Raoufinia R, Afrasiabi P, Dehghanpour A, Memarpour S, Hosseinian SHS, Saburi E, Naghipoor K, Rezaei S, Haghmoradi M, **Keyhanvar N**, et al. The landscape of microRNAs in bone tumor: a comprehensive review in recent studies. *MicroRNA*. 2024;13(3):175–201.
9. Sarvari R, Keyhanvar P, Agbolaghi S, Roshangar L, Bahremani E, **Keyhanvar N**, Haghdoost M, Keshel SH, Taghikhani A, Firouzi N, et al. A comprehensive review on methods for promotion of mechanical features and biodegradation rate in amniotic membrane scaffolds. *Journal of Materials Science: Materials in Medicine*. 2022;33(3):32.
10. Hajipour H, Farzadi L, Latifi Z, **Keyhanvar N**, Navali N, Fattahi A, Nouri M, Dittrich R. An update on platelet-rich plasma (PRP) therapy in endometrium and ovary related infertilities: clinical and molecular aspects. *Systems Biology in Reproductive Medicine*. 2021;67(3):177–188.
11. Valizadeh A, Asghari S, Bastani S, Sarvari R, **Keyhanvar N**, Razin SJ, Khiabani AY, Yousefi B, Yousefi M, Shoae-Hassani A, et al. Will stem cells from fat and growth factors from blood bring new hope to female patients with reproductive disorders? *Reproductive Biology*. 2021;21(2):100472.
12. **Keyhanvar N**, Zarghami N, Bleisinger N, Hajipour H, Fattahi A, Nouri M, Dittrich R. Cell-based endometrial regeneration: current status and future perspectives. *Cell and Tissue Research*. 2021;384(2):241–254.
13. **Keyhanvar N**, Zarghami N, Seifalian A, Keyhanvar P, Sarvari R, Salehi R, Rahbarghazi R, Ranjkesh M, Akbarzadeh M, Mahdipour M, et al. The combined thermoresponsive cell-imprinted substrate, induced differentiation, and "KLC Sheet" formation. *Advanced Pharmaceutical Bulletin*. 2021;12(2):356.
14. Raoufinia R, Balkani S, **Keyhanvar N**, Mahdavipor B, Abdolalizadeh J. Human albumin purification: a modified and concise method. *Journal of Immunoassay and Immunochemistry*. 2018;39(6):687–695.
15. Raoufinia R, Mahdavipor B, **Keyhanvar N**, Abdolalizadeh J, et al. A modified and concise method for albumin purification. 2017.
16. Raoufinia R, Mota A, **Keyhanvar N**, Safari F, Shamekhi S, Abdolalizadeh J. Overview of albumin and its purification methods. *Advanced Pharmaceutical Bulletin*. 2016;6(4):495.
17. **Keyhanvar N**, Amanpour R, Nouri M. A substrate for stem cell culture and differentiation. *Journal of Skin and Stem Cell*. 2016;3(3):e64047.
18. Yazdani Y, **Keyhanvar N**, Tabaraei A. Cloning and functional assessment of recombinant human hepcidin-25 in the baculovirus expression system. *Brazilian Archives of Biology and Technology*. 2014;58(1):90–95.
19. Yazdani Y, **Keyhanvar N**, Kalhor HR, Rezaei A. Functional analyses of recombinant mouse hepcidin-1 in cell culture and animal model. *Biotechnology Letters*. 2013;35(8):1191–1197.
20. **Keyhanvar N**, Tabarraei A, Yazdani Y. Production of recombinant bacmid containing the coding sequence of human hepcidin. *Medical Laboratory Journal*. 2013;7(2):1–6.

TEACHING & MENTORING

- Mentored a research associate through an 18-month training period leading to five graduate school interviews and PhD program admission (UCSF, 2025–present).
- Supervised an undergraduate researcher and a rotating PhD student in experimental design and hands-on laboratory technique (UCSF, 2025–present).
- Mentored a research technician through daily scientific guidance, CV development, interview coaching, and literature review, supporting a successful graduate school application (UCSF, 2023–2025).
- Trained and mentored 15+ graduate and undergraduate students in experimental design, laboratory technique, data analysis, and scientific communication (Tabriz, 2014–2023).

PRESENTATIONS

Oral Presentation

- The Interplay Between Mechanotransduction and Immunosenescence: Insights into Aging, Bakar Aging Research Institute (BARI) Symposium, San Francisco, CA, USA (Apr 2025)
- A novel scaffold-free tissue engineering method for skin regeneration, The 2nd National Festival and International Congress on Stem Cell and Regenerative Medicine, Tehran, Iran (Jul 2017)
- Cell-imprinted Plastic compression as a novel method for topographically guide the differentiation of stem cells, The First National and International Congress on Stem Cell and Regenerative Medicine, Tehran, Iran (May 2016)
- Production of a novel construct for expression of the recombinant human hepcidin-25, 1st National Congress on Biotechnology, Golestan, Iran. Keyhanvar, N., Tabarrae, A., Yazdani, Y. (July 2012)

Poster Presentation

- Keyhanvar, N., Zhurikhina, A., Biancardi, F., Martens, H., YAP/TAZ Mechanotransduction, a Key Player of The Aging Tissue, Bakar Aging Research Institute (BARI) Symposium, San Francisco, CA, USA (Apr 2025)
- Keyhanvar, N., Zarghami, N. (20-22 May, 2015) "Cell Sheet Engineering", A Novel Scaffoldfree Stem cell-based Approach In Tissue Engineering And Regenerative Medicine, International Congress on Stem Cells and Regenerative Medicine in Mashhad, Iran.
- Keyhanvar, N., Yazdani, Y. (16-19 April, 2013) Expression of the key iron regulatory hormone, human hepcidin, in baculovirus expression system, 13th Iranian congress of Biochemistry & 5th International congress of Biochemistry and molecular Biology, Yazd, Iran.
- Keyhanvar, N., Yazdani, Y. (15-18 January, 2013) Production of the functional part of human iron hormone in baculovirus expression system, The 5th International Congress of Laboratory & Clinic "Blood Disorders" Iranian Scientific Association of Clinical Laboratory, Tehran, Iran.
- Keyhanvar, N., Yazdani, Y. (9-12 October, 2012) Production of e novel recombinant bacmid containing the functional sequence of human hepcidin-25, 13th Annual Research Congress of Iran`s Medical Sciences Students, Babol, Iran.
- Yazdani, Y., Keyhanvar, N., Rezaei, A. (15-18 January, 2013) Regulatory T cell response is increased during fungal infections, The 5th International Congress of Laboratory & Clinic "Blood Disorders" Iranian Scientific Association of Clinical Laboratory, Tehran, Iran.
- Raoufinia, R., Balkani, S., Mahdavipour, B., Keyhanvar, N., Abdolalizadeh, J., (23-24 Feb, 2017) A Modified and Concise Method for Albumin Purification, Third Annual Research Conference of the Semnan University of Medical Sciences, Semnan, Iran.

GRANTS, AWARDS AND HONORS

- Selected for oral presentation – Bakar Aging Research Institute (BARI) Symposium, San Francisco, US. 2025
- Volunteer – 50th RESI Conference, J.P. Morgan Healthcare Week, San Francisco, US. 2025
- Invited peer reviewer – BioImpacts journal (6 manuscripts), Iran. 2023
- 3rd Prize – Technology, Entrepreneurship & Research Festival (FAKT), Iran. 2016
- Competitive research grant award – Stem Cell & Regenerative Medicine Institute, Iran. 2014
- Top GPA, Distinguished Student - Medical Biotechnology Program. 2013
- 1st Prize, Best Oral Presentation & Best Abstract – National Congress on Biotechnology, Iran. 2012

PROFESSIONAL SERVICE

- Invited Peer Reviewer, *BioImpacts* journal (6 manuscripts), 2023
- Volunteer, 50th RESI Conference, J.P. Morgan Healthcare Week, San Francisco, CA, 2025

REFERENCES

- **Dr. Hanna Martens** (Hanna.martens@ucsf.edu), Assistant Professor, Department of Biochemistry & Biochemistry, University of California San Francisco (UCSF), San Francisco, CA, USA.
- **Dr. Caroline Vissers** (caroline.vissers@ucsf.edu), Former Sandler Faculty Fellow, Principal Investigator at Vissers lab, University of California San Francisco (UCSF), San Francisco, CA, USA.
- **Dr. Hani Goodarzi** (hani.goodarzi@ucsf.edu), Associate Professor, Department of Biophysics & Biochemistry, University of California San Francisco (UCSF), San Francisco, CA, USA.